Introduction to Bioinformatics

Variant Calling

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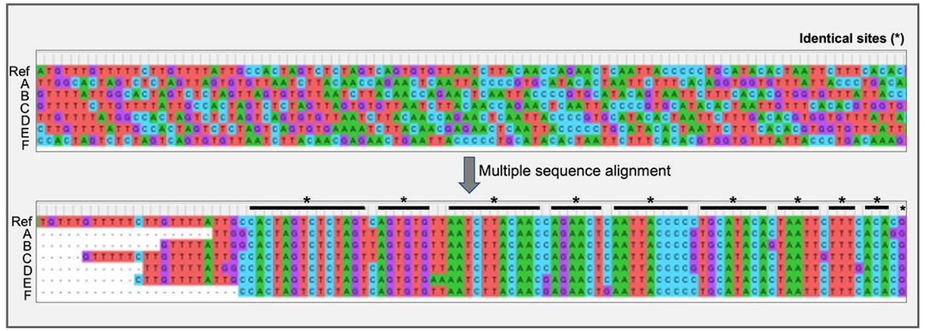
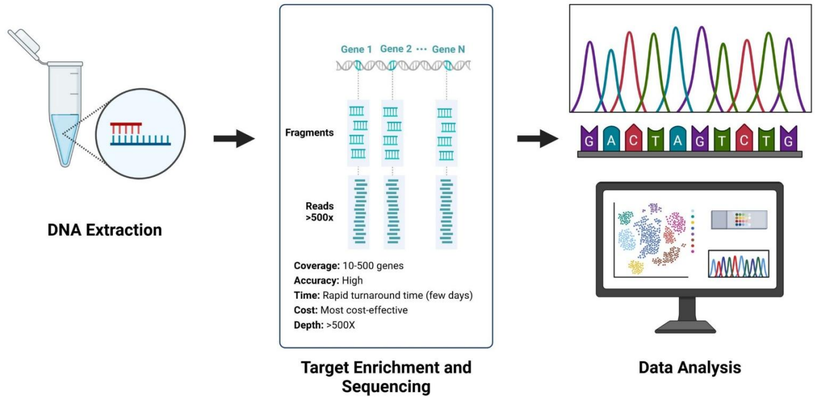
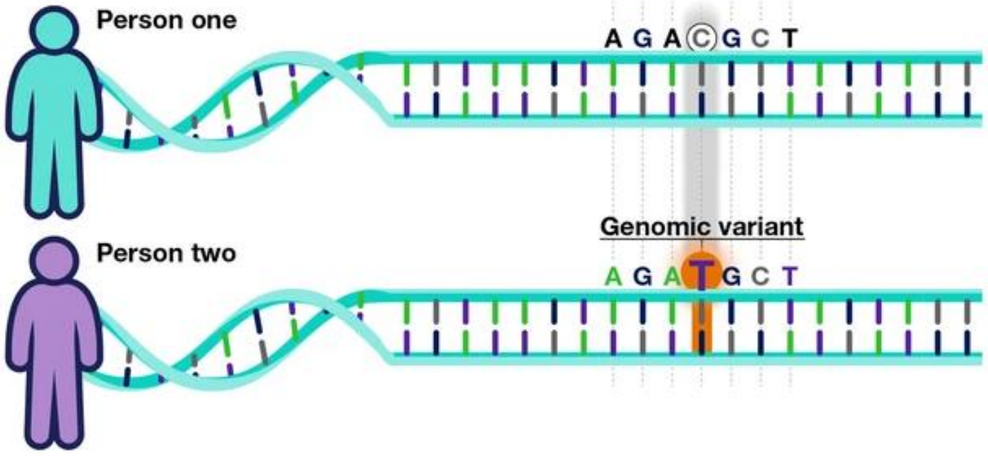
Basic Workflow



Sequencing → Alignment → Variant Calling





Types of Variants

What Types of Variants Are Detected?

What are we looking for?

- **<u>Variant</u> Calling** = is the bioinformatics process of identifying differences between sequenced individual genome and a known reference genome.
- A **Variant** = is any genetic variation found

Main sources of variants:

- **DNA replication errors**→ mistakes during cell division (e.g., wrong base inserted)
- **DNA damage**→ caused by UV radiation, chemicals, or environmental factors
- **Defective DNA repair**→ errors are not corrected properly
- **Chromosomal rearrangements**→ large-scale changes such as translocations or inversions

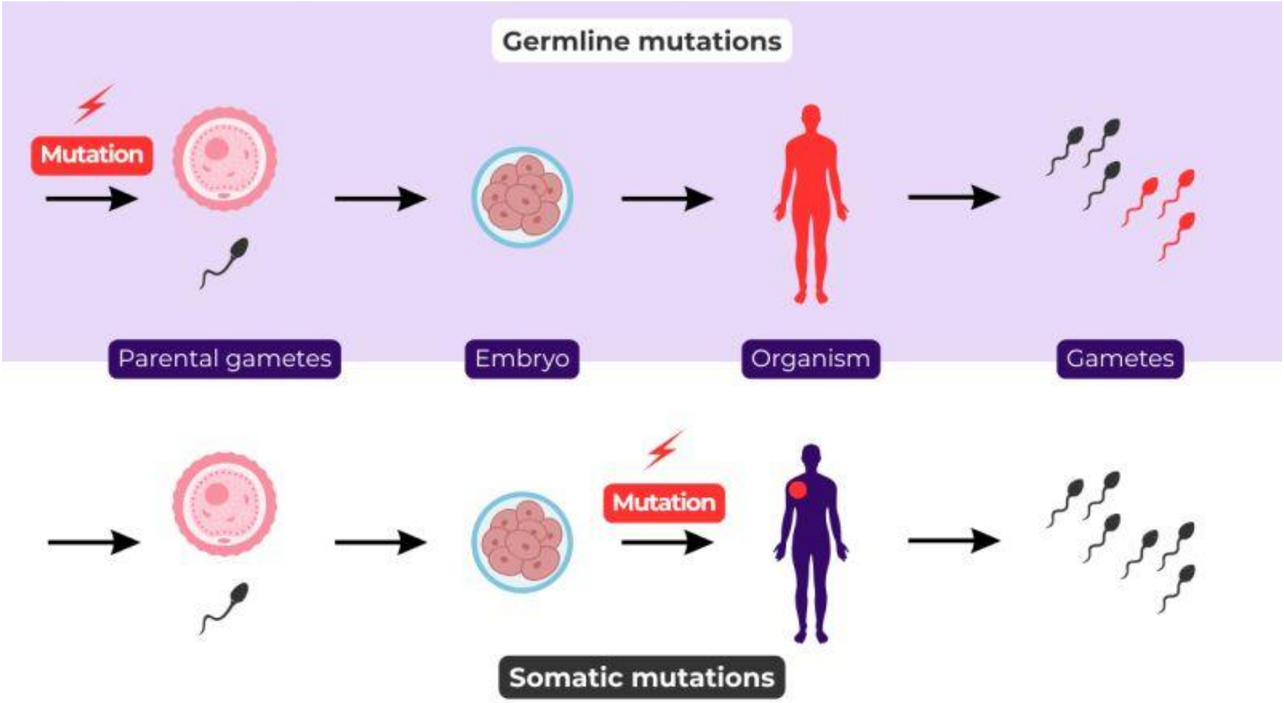
Types of variants by origin

- **Germline variants**

  - inherited, present in all cell e.g. genetic diseases

- **Somatic variants**

  - arise during life, present only in some cells e.g., cancer



What are we looking for?

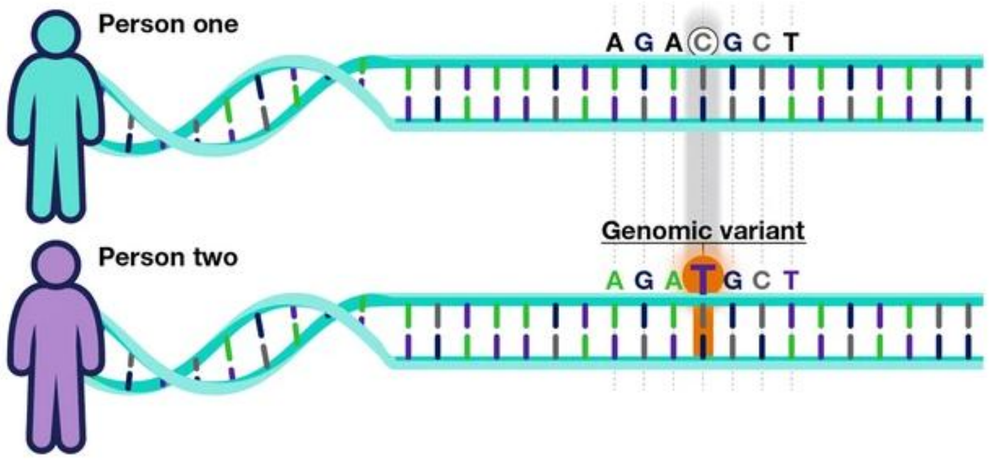
- Types:

  - **SNVs** (SNPs)

  - **Indels**

  - **Structural Variants**

  - **Copy Number Variations**



- Variant represents a specific, localized change in the DNA sequence compared to the reference.

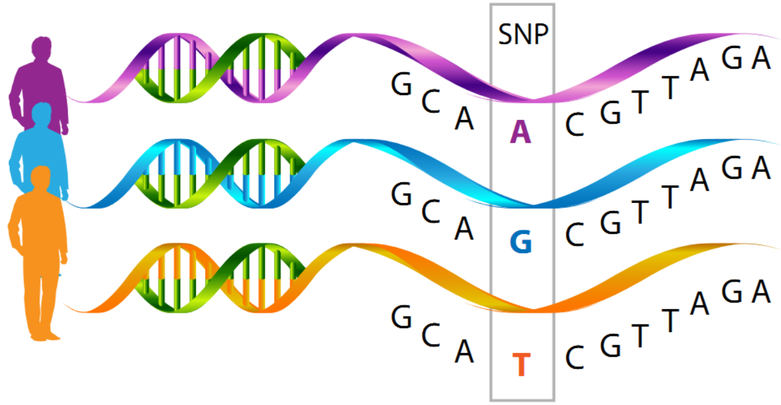
SNVs

- **S**ingle **N**ucleotide **V**ariants
- **Definition**: SNVs are alterations in DNA where a <u>single nucleotide</u> (A, T, C, or G) is substituted, inserted, or deleted at a specific genomic position.
- They are the most common type of genetic variation, occurring in both coding and non-coding regions, and can arise from DNA damage, replication errors, or chemical exposure.
- **<u>SNV vs. SNP</u>**: SNP is a type of SNV that is present in at least 1% of the population. If a variant is rare (found in less than 1% of the population), it is generally called an SNV rather than a SNP.

Single Nucleotide Variation

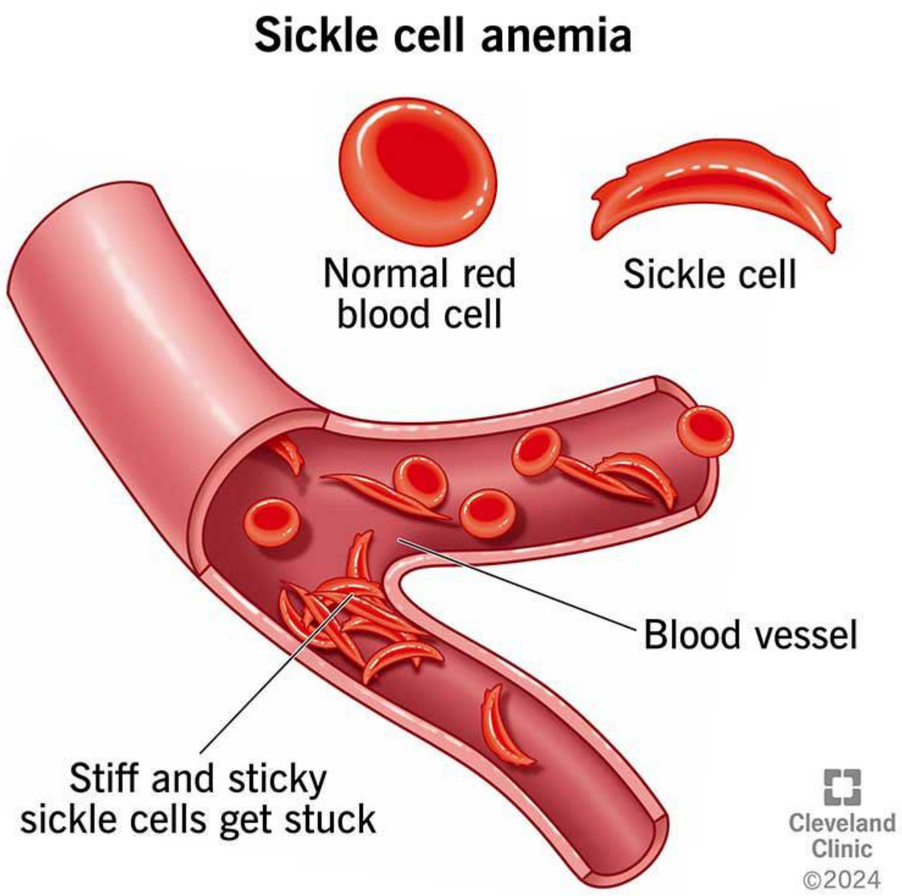
SNPs

- **S**ingle **N**ucleotide **P**olymorphism
- **Definition:** A single base-pair substitution (e.g., C instead of T) that must exist in at least 1% of the population to be considered a SNP**.**
- **Frequency & Location:** There are millions of SNPs in the human genome, occurring in both coding (genes) and non-coding regions.
- **Significance:** While most SNPs do not affect health, some are associated with diseases and influence how individuals react to drugs or environmental factors.



SNPs - Examples

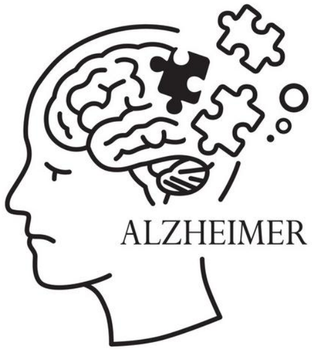
- **<u>Sickle cell disease</u>** *(Kosáčikovitá anémia)*
- **gene**: *HBB* (hemoglobin subunit beta)- chr11
- **SNP**: A → T (GAG → GTG)
- **Amino acid**: glutamate → valine (position 6)

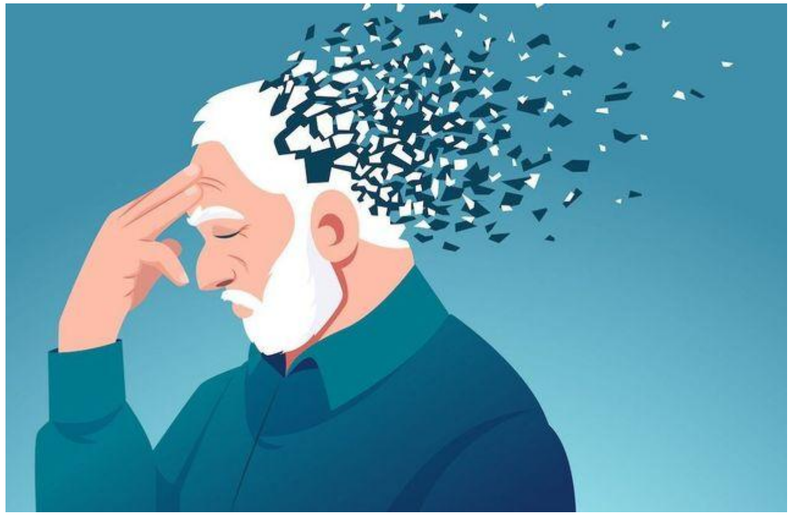


- This small SNP causes a change in the structure of hemoglobin, resulting in red blood cells being deformed into a sickle shape

https://www.ncbi.nlm.nih.gov/snp/rs334#frequency_tab

SNPs - Examples





- **<u>Alzheimer's disease</u>**
- **gene**: APOE (apolipoprotein E) – chr19
- **SNP**: C → T (rs429358), C → T (rs7412)
- **Amino acid**: Cys → Arg (position 112), Arg → Cys (position 158)

- This combination of SNPs determines APOE variants (ε2, ε3, ε4).
- The ε4 variant is associated with an increased risk of Alzheimer's disease because it impairs the clearance of β-amyloid in the brain.

SNPs – Cancer Examples

- SNPs can alter gene function or regulation, contributing to cancer development
- they may lead to:

  - changes in protein structure

  - increased or decreased gene expression (DNA→RNA → Protein)

Examples:

- Breast cancer (FGFR2)→ SNPs increase gene expression → promotes tumor growth
- Lung cancer (CHRNA3)→ affects receptor function → linked to smoking-related risk
- Melanoma (SLC45A2)→ amino acid change → altered pigmentation protein

Indels

- **In**sertions and **Del**etion**s**
- is a type of genetic mutation that occurs:

  insertion - insertion of one or more nucleotides into DNA

  deletion - removal of one or more nucleotides
- INDEL can cause:

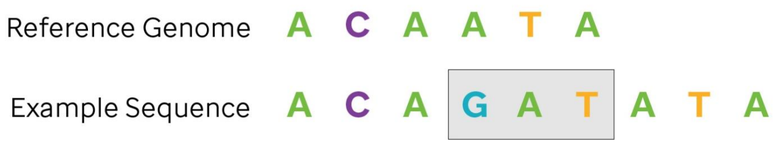
  - *frameshift mutation* - reading frame shift (if number of nucleotides ≠ multiple of 3)

  - *amino acid change*

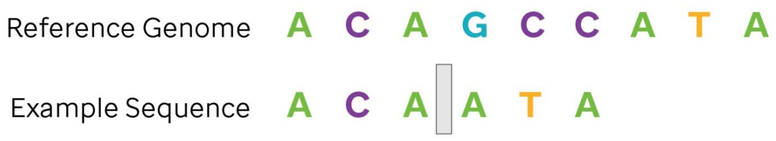
  - *premature stop codon*

  - *non-functional protein*
- Such mutations often cause more severe disease than SNPs because they change a large part of the protein.

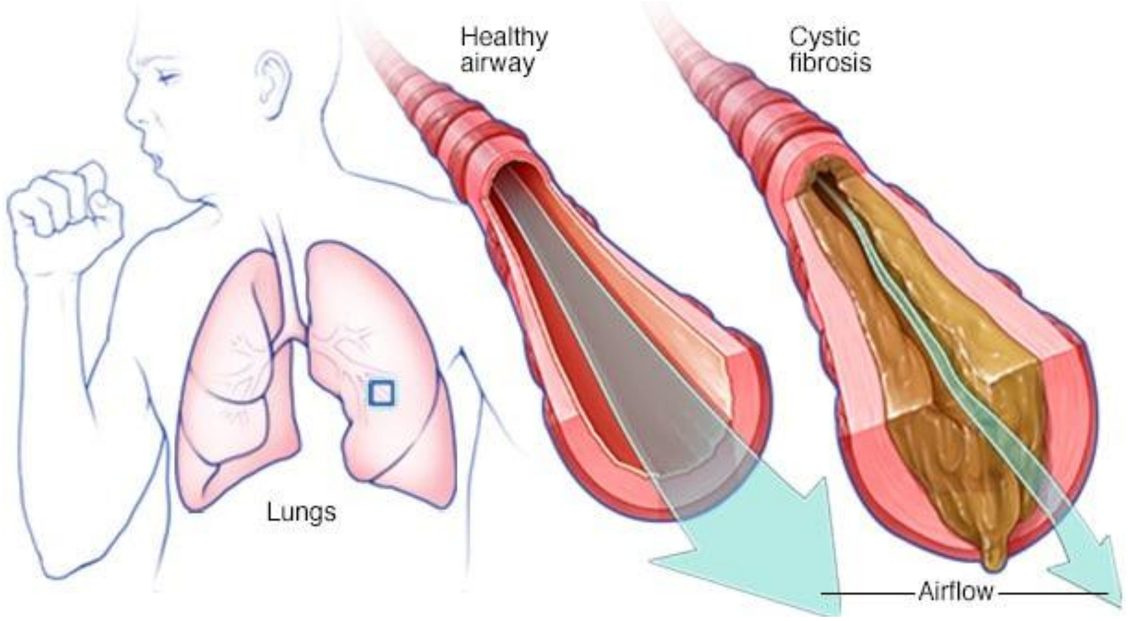
Indel: Insertion



Indel: Deletion



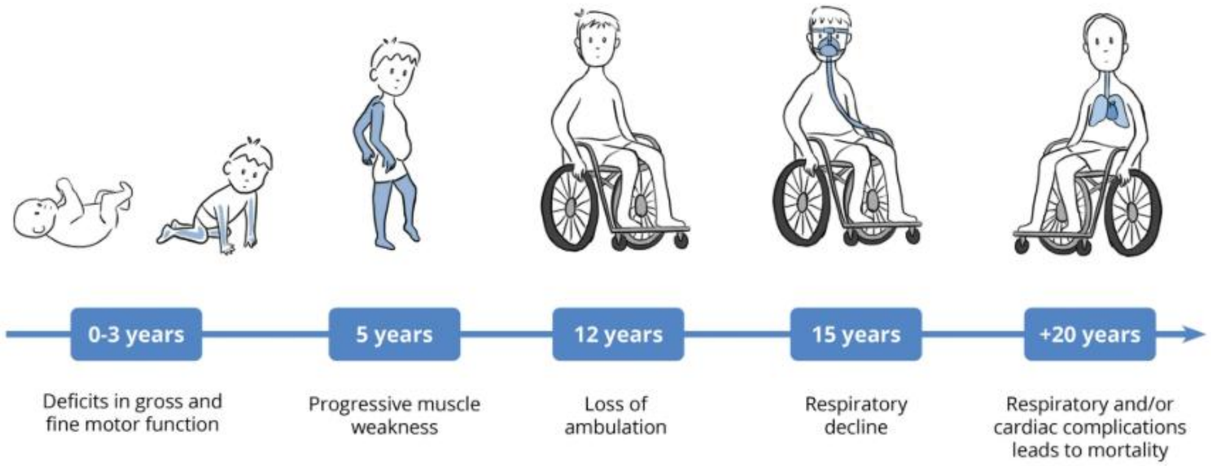
Indels - Example



- **<u>Cystic fibrosis</u>**
- **gene**: CFTR (cystic fibrosis transmembrane conductance regulator) – chr7
- **INDEL**: deletion of 3 nucleotides (ΔF508)
- **Amino acid:** F508 phenylalanine deleted (position 508)

- This small deletion removes one amino acid (phenylalanine), causing the CFTR protein to misfold. As a result, the protein is degraded and cannot function properly, leading to thick mucus buildup in lungs and other organs.

Indels - Example



- **<u>Duchenne muscular dystrophy</u>**
- **gene**: DMD– chrX
- **INDEL**: deletion of one or more exons
- **Amino acid:** disrupted reading frame → no functional **dystrophin**

- This deletion causes a **frameshift mutation**, leading to production of a truncated and non-functional dystrophin protein. As a result, muscle fibers are damaged over time, leading to progressive muscle weakness and degeneration.

Indels - Causes

- Indels can disrupt the reading frame of a gene, leading to major changes in protein structure
- they may cause:

  - **Frameshift mutations**→ insertion or deletion not divisible by 3 changes the reading frame→ completely different amino acid sequence

  - **Premature stop codons**→ mutation introduces an early STOP signal→ protein is shortened

  - **Loss of protein function**→ protein cannot fold properly or perform its role→ often affects tumor suppressor genes

Mutations

- *frameshift mutation* - reading frame shift (if number of nucleotides ≠ multiple of 3)

- AAT CCA TGA …. → AAU CCA UGA … → Asparagine(Asn) – Proline (Pro) – STOP codon

- THE CAT ATE THE RAT
- Remove letter on second position: T**H**E CAT ATE THE RAT → TEC ATA TET HER AT

SVs

- **S**tructural **V**ariants
- large-scale genomic alterations typically **>50 base pairs** (bp) in size, involving **deletions**, **duplications**, **inversions**, or **translocations** of DNA segments
- Unlike SNPs or small indels, they can affect large parts of genes or entire genes.
- Types of SVs:

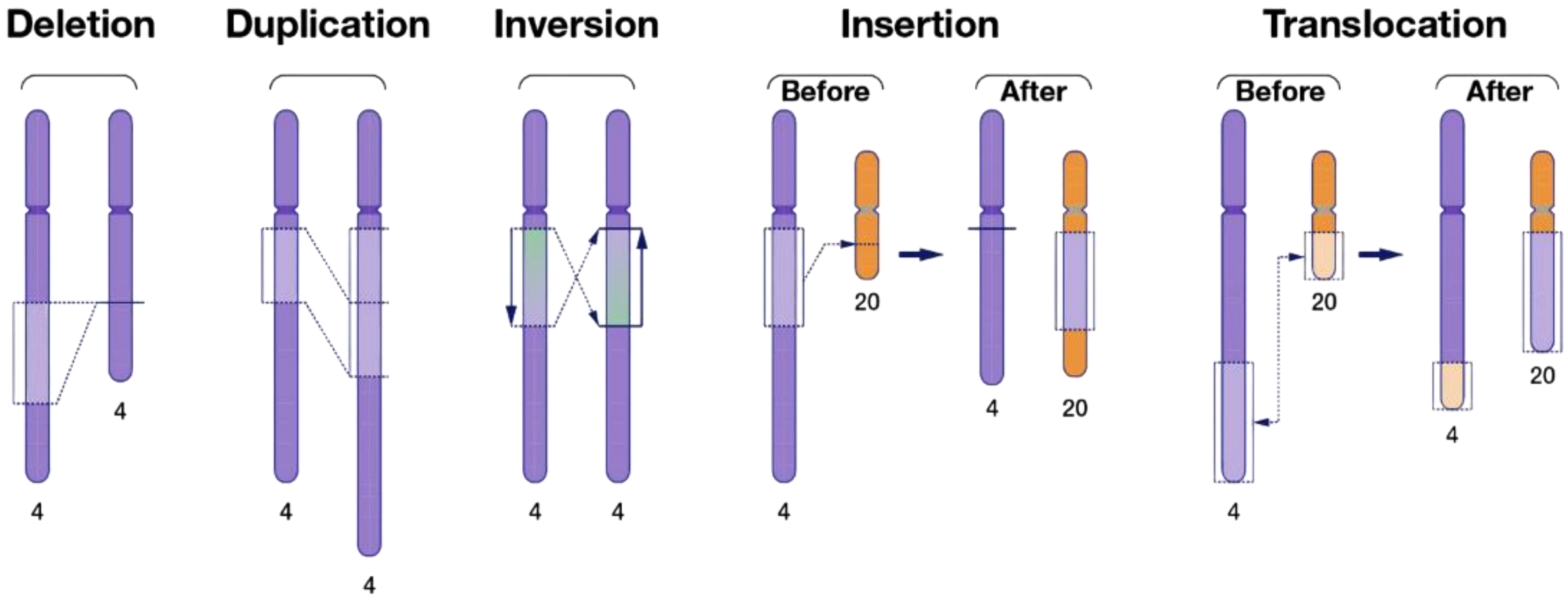
  - **Deletion** = deletion of a larger segment of DNA

  - **Duplications/Insertions** = duplication/insertion of a DNA segment

  - **Inversions** = rotation(inversion) of a segment of DNA in a chromosome

  - **Translocations** = transfer of a DNA segment to another chromosome

SVs


Deletion
Duplication
Inversion
Insertion
Translocation
Before
After
Before
After
4
4
4
4
4
4
4
20
4
20
20
4
20
4

SVs

- SV can cause :

  - *delete the entire gene*

    → large deletion removes a whole gene→ protein is not produced at all

  - *alter gene regulation*

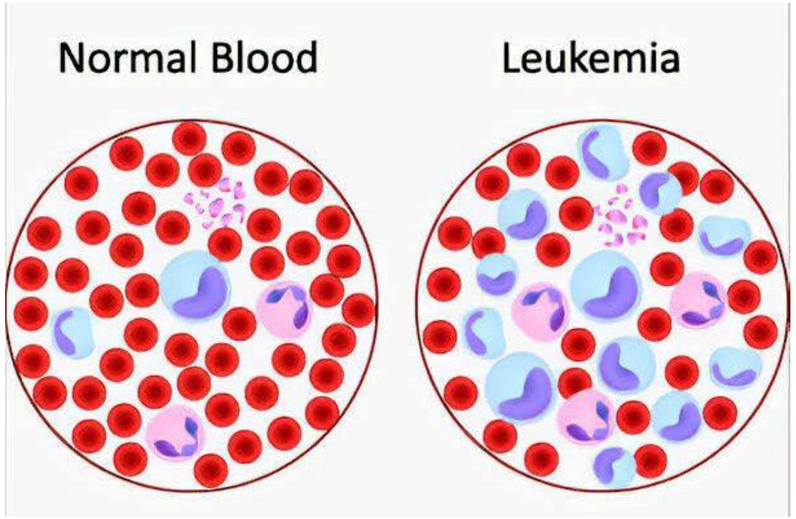
    → variant changes how a gene is expressed→ may be expressed too much or too little

  - *create a fusion gene*

    → parts of two different genes are joined together→ new abnormal protein is produced

    → often seen in cancer

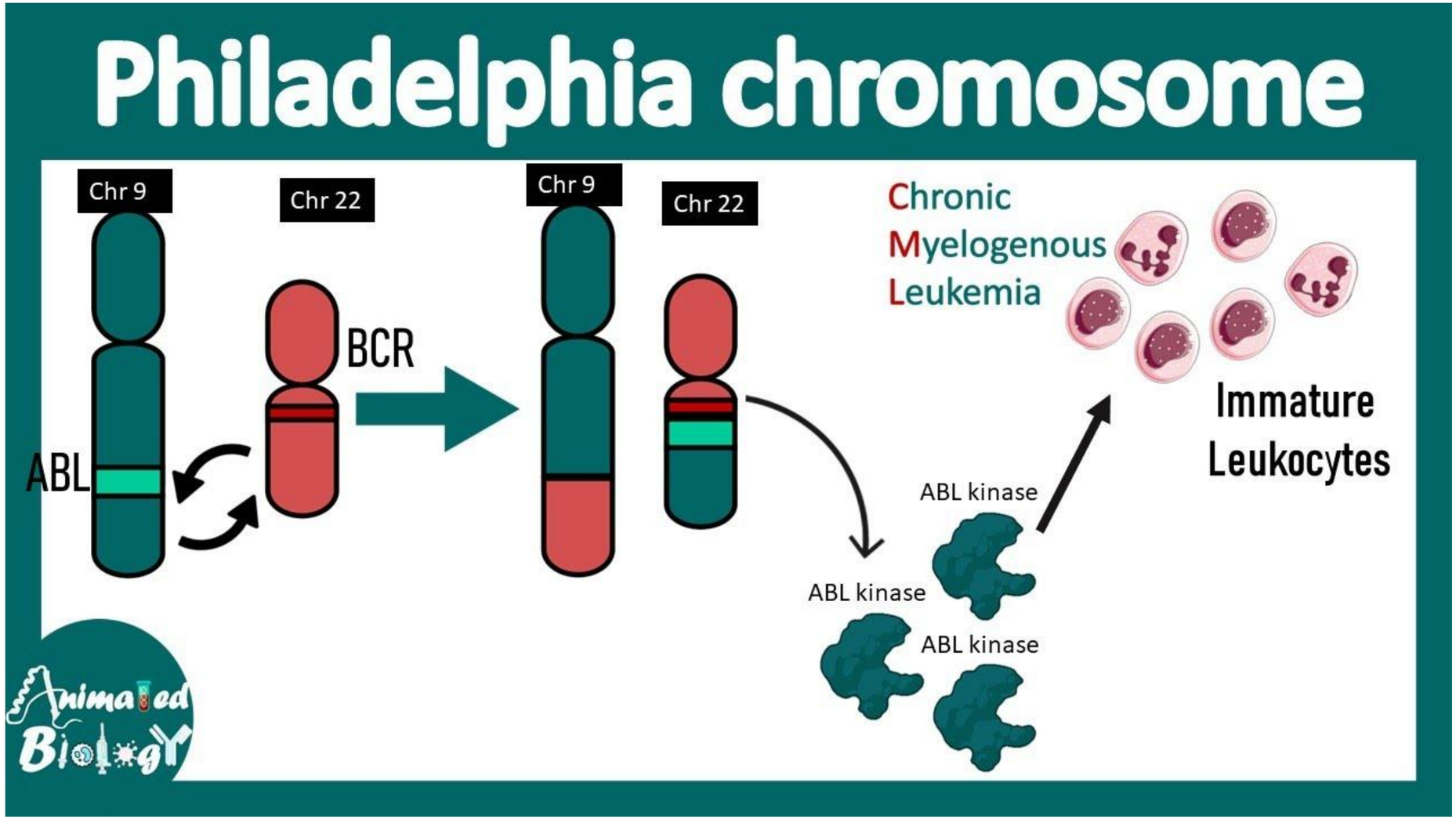
SVs - Examples



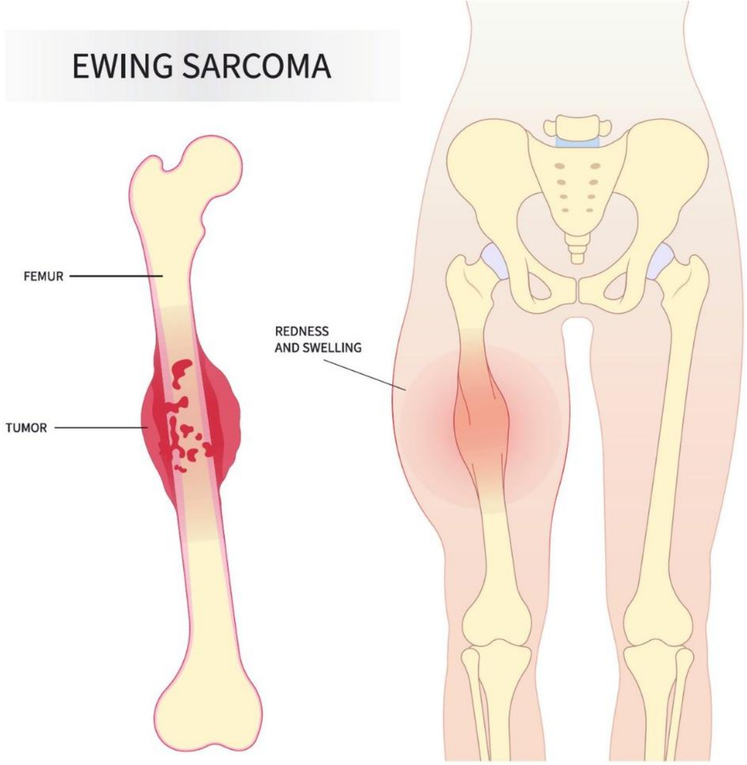
- **<u>Myeloid leukemia</u>**
- **gene**: BCR (chr22) / ABL (chr9)
- **SV**: translocation t(9;22) – Philadelphia chromosome
- **Fusion protein**: BCR-ABL1 (constitutively active tyrosine kinase)

- This structural variant creates a fusion gene (BCR-ABL1), which produces a constantly active tyrosine kinase. This leads to uncontrolled cell division and proliferation of myeloid cells, causing leukemia.

SVs - Examples



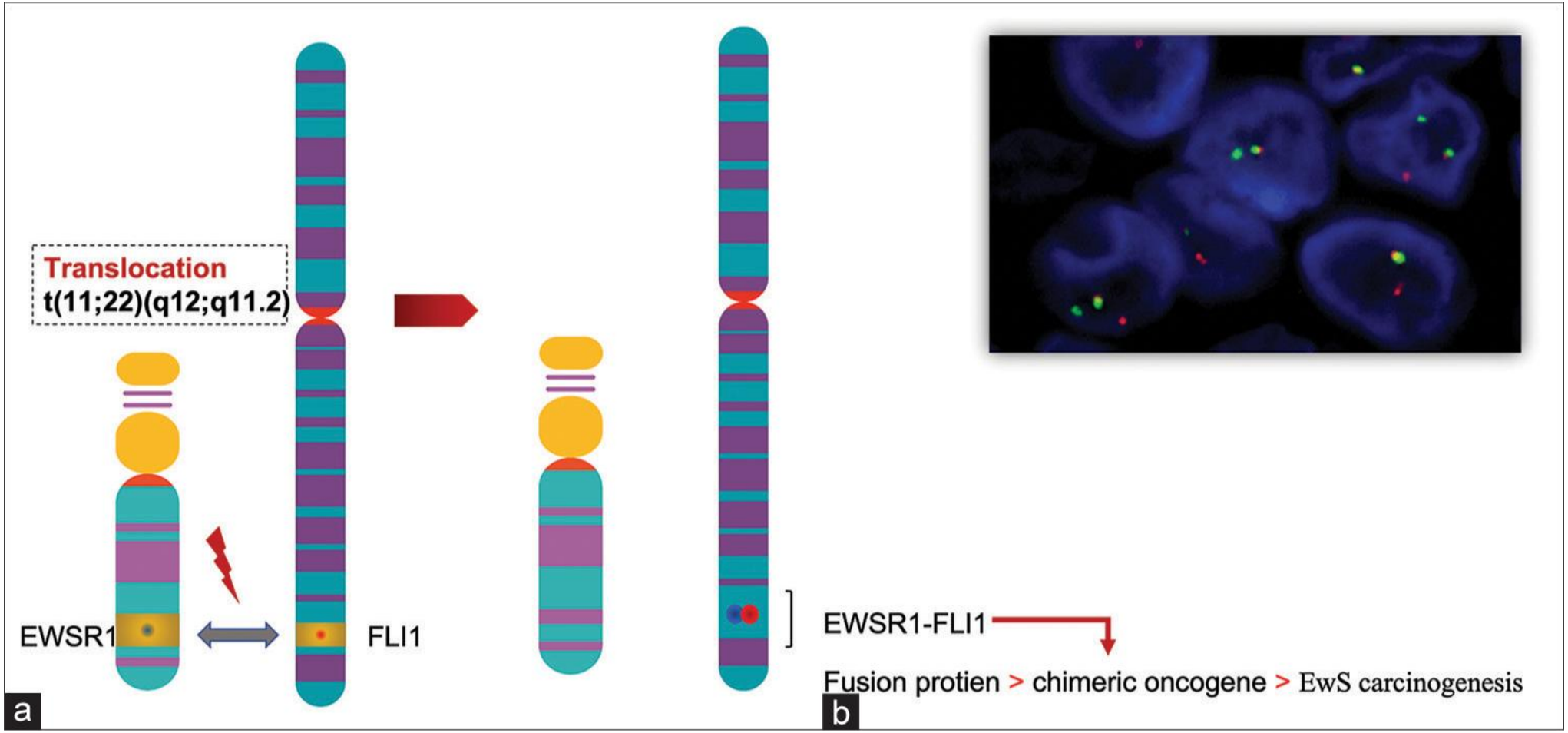
SVs - Examples



- **<u>Ewings sarcoma</u>**
- **gene**: EWSR1 (chr22) / FLI1 (chr11)
- **SV**: translocation t(11;22)
- **Fusion protein**: EWSR1-FLI1 (aberrant transcription factor)

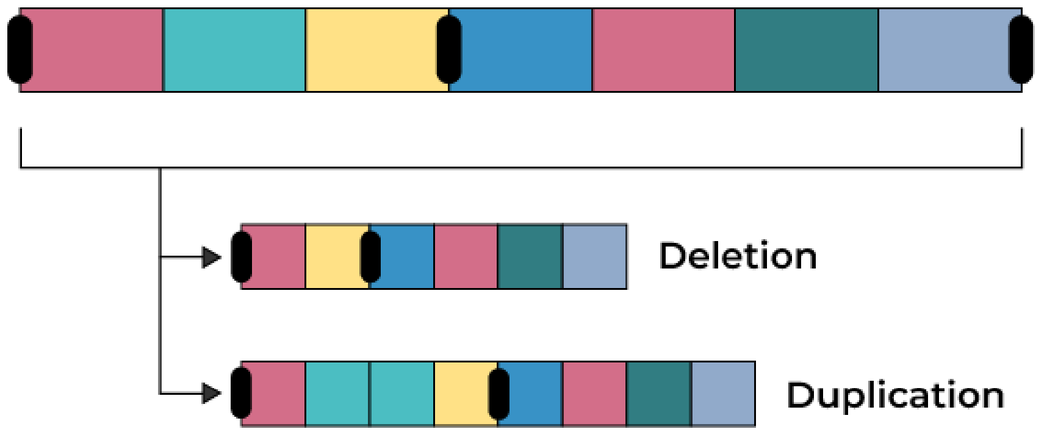
- This structural variant creates a fusion gene (EWSR1-FLI1), which acts as an abnormal transcription factor. It alters gene expression and drives uncontrolled cell growth, leading to the development of Ewing's sarcoma.

SVs - Examples



Copy Number Variants - CNVs

- genomic alterations where the number of copies of a DNA segment changes, including deletions (loss) or duplications/amplifications (gain) typically involve larger DNA regions (from kilobases to megabases)
- **Copy Number Loss:** This means that a section of DNA is missing. If one copy is missing, it is called a deletion, which can result in the body missing important instructions for making proteins.
- **Copy Number Gain:** This means that a section of DNA is present in the genome more times than normal. This phenomenon is called duplication (or amplification) and can cause a cell to produce too much of a particular protein.



Importance of Variant Calling

Why is important?

Importance

- Without variant calling, we would just have a bunch of data.

- Thanks to it, we can:
  Diagnose rare diseases→ identify disease-causing mutations
  Oncology→ find mutations for targeted therapy
  Personalized medicine→ tailor treatment to individual patients
  Pharmacogenomics→ predict drug response
  Infectious diseases→ track pathogen evolution (e.g., viruses)
  Population genetics→ study human evolution and migration

Variant Calling

Tools and Formats

How does Variant calling works

Variant Callers

How do variant calling tools work?

1. **Input data** BAM file+ reference genome

2. **Compare reads to reference**

- tool checks each position in the genome and looks for differences (A vs T, insertions, deletions)

3. **Collect evidence**

- how many reads support REF vs ALT
- sequencing quality + read depth (coverage)

4. **Decide if variant is real**

- statistical model (is it error or real mutation? Is sequenator error?)
- applies filters (quality thresholds)

5. **Output list of variants in VCF file**

How do variant calling tools work?

4. Decide if variant is real

- **Binomial distribution** → helping decide if the signal is random or real
- **Likelihood** → calculates how well different genotypes (0/0, 0/1, 1/1) explain the observed data and selects the most likely one
- **Bayesian approach** → combines observed data with prior expectations (e.g. variant frequency) to estimate the probability of each genotype
- **Phred scale** → converts error probabilities into a log scale, making it easier to compare quality scores across bases, reads, and variants
- **Fisher's exact test** → testing whether alternative alleles are unevenly distributed across forward and reverse

How do variant calling tools work?

```
Reference:  A A A A A
Reads:      A A A A A
            A A G A A
            A A G A A
            A A A A A
            A A G A A
```

→ possible variant: A → G
→ support: 3/5 reads (60%)
→ probably the real variant

Variant Callers

Which tool to use and when?

Small Variants (Snps + Indels)

Tools for small variants (<50 bp)

- **GATK HaplotypeCaller**

→ *gold standard for germline SNPs and indels*

- **Strelka2**

→ *fast and accurate for both germline and somatic variants*

- **FreeBayes**

→ *simple and flexible open-source alternative*

- **DeepVariant**

→ *very high accuracy, especially for long-read data*

Somatic (cancer)

Tools for somatic variants

- **GATK Mutect2**

→ *best for detecting mutations in tumor samples*

- **Strelka2**

→ *works well for tumor-normal comparisons*

- **Tumor vs Normal sample**

  Tumor sample = DNA from cancer cells

  Normal sample = DNA from healthy tissue of the same patient

  comparing them helps identify somatic mutations specific to the tumor

Structural variants

Tools for structural variants (>50 bp)

- **Manta**

→ *fast and widely used for structural variants*

- **Delly**

→ *good for complex rearrangements*

VCF format

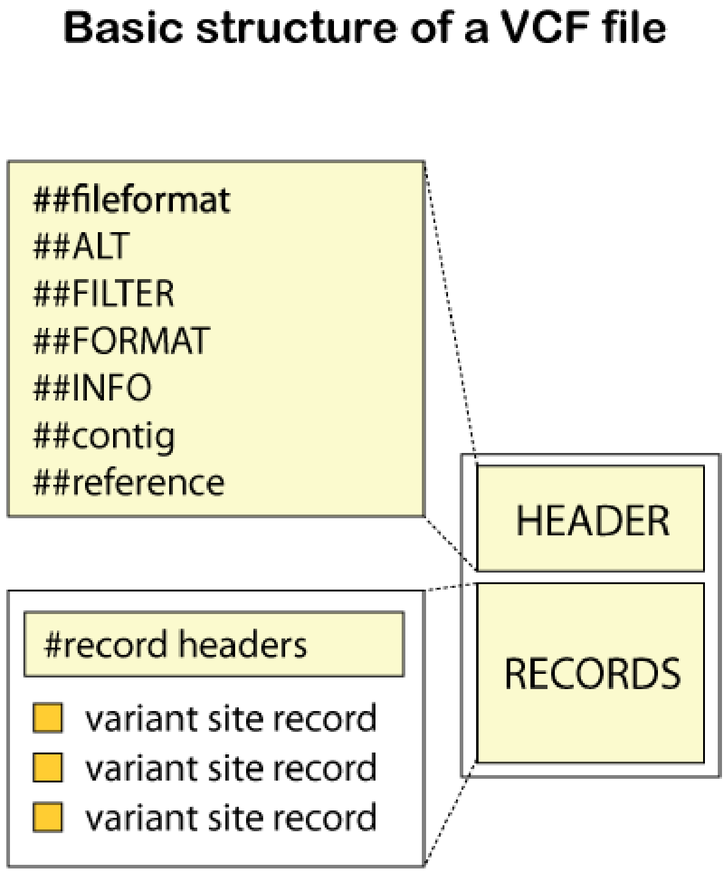
Variant Call Format

VCF File

- **V**ariant **C**all **F**ormat – **VCF**
- *.vcf*
- A VCF (Variant Call Format) file is a **standard text file format** used in bioinformatics to store **genetic variants detected** during variant calling.
- These variants include SNPs, Insertions and deletions (indels), Structural variants

Structure of VCF file

- VCF is a common output format for variant calling programs due to its relative simplicity and scalability.
- **VCF file is composed of two main parts:**
  - Header
  - Variant Call Records



Structure of VCF file

- <u>Header</u>

- The header starts with ## and contains **metadata** about the file.
- Example:

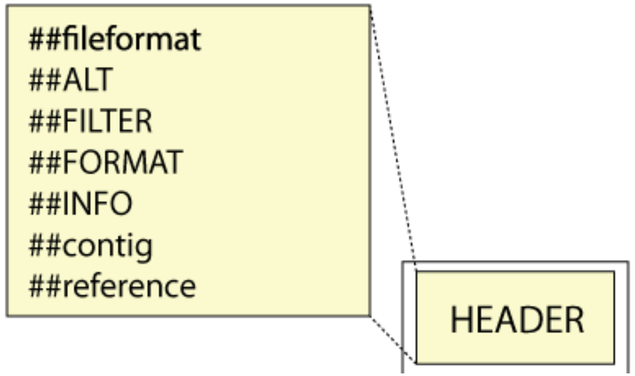
```
##fileformat=VCFv4.2

##source=GATK

##reference=GRCh38

##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">

##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
```

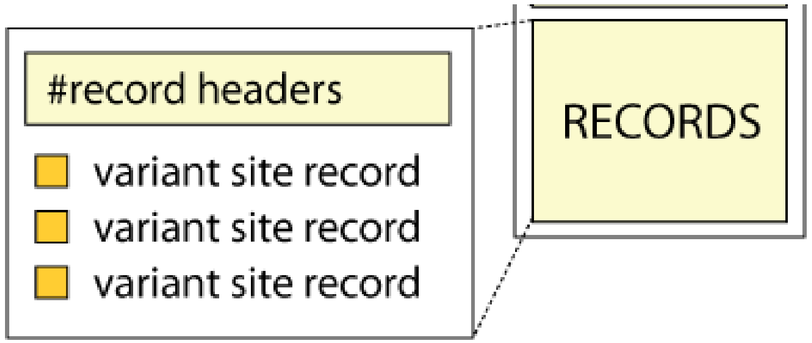


Structure of VCF file

- Variant Call Records

- This line starts with a single # and defines the columns.
- Example:

```
#CHROM POS ID REF ALT QUAL FILTER INFO
```



Structure of VCF file

Column	Description
#CHROM	Chromosome - On which chromosome is the variant located
POS	Position of the variant – the exact coordinate on chromosome
ID	Variant identifier (e.g., rsID from dbSNP database)
REF	Reference allele - What base (A, C, T, G) is in the reference genome.
ALT	Alternative allele - What base was found in the patient.
QUAL	Quality score of the variant - the certainty that it is not a sequencing error.
FILTER	Filter status (PASS if variant passed filters)
INFO	Additional data - (e.g., read depth, frequency in the population).

Example VCF

Example

VCF header

```
##fileformat=VCFv4.0
##fileDate=20100707
##source=VCFtools
##reference=NCBI36
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)">
##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
```

Body

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1	SAMPLE2
1	1	.	ACG	A,AT	.	PASS	.	GT:DP	1/2:13	0/0:29
1	2	rs1	C	T,CT	.	PASS	H2;AA=T	GT:GQ	0|1:100	2/2:70
1	5	.	A	G	.	PASS	.	GT:GQ	1|0:77	1/1:95
1	100	.	T	<DEL>	.	PASS	SVTYPE=DEL;END=300	GT:GQ:DP	1/1:12:3	0/0:20

Mandatory header lines

Optional header lines (meta-data about the annotations in the VCF body)

Reference alleles (GT=0)

Alternate alleles (GT>0 is an index to the ALT column)

Phased data (G and C above are on the same chromosome)

Deletion

SNP

Large SV

Insertion

Other event

Variant Genotypes

- Everyone has two alleles at every genomic position→ one inherited from the mother and one from the father
- Homozygous reference (0/0)→ both alleles are the same as the reference genome
- Heterozygous (0/1)→ one allele is reference, the other is alternative
- Homozygous alternate (1/1)→ both alleles differ from the reference

Example

Reference base: A

From mother: A

From father: G

Genotype: A/G → heterozygous (0/1)

Example VCF

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1
1	1	.	ACG	A,AT	.	PASS	.	GT:DP	1/2:13
1	2	rs1	C	T,CT	.	PASS	H2;AA=T	GT:GQ	0/1:100
1	5	.	A	G	.	PASS	.	GT:GQ	1/0:77
1	100	.	T	<DEL>	.	PASS	SVTYPE=DEL;END=300	GT:GQ:DP	1/1:12:3

Heterozygous

Homozygous alternate

Heterozygous – multialellic variant

REF = A
ALT = C,G

Why VCF is Important

- VCF files enable:
  - Standardized storage of genetic variation
  - Easy sharing and analysis of variants
  - Compatibility with many genomic analysis tools
  - Efficient filtering and annotation of variants

Questions?

Thank you for your attention